

LSE DATA ANALYTICS CAREER ACCELERATOR · EMPLOYER PROJECT

# VP ANALYTICS

Location Health Index

## Final Technical Report

PREPARED BY

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Employer project engagement: **VP Analytics**

Submission Date **18 September 2026**

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*Ordered so the conclusion comes before the evidence, and the evidence before the method.*

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01

## Executive Summary

The Location Health Index (LHI) is a reproducible, evidence-tested digital-health score for 19,154 quick-service-restaurant locations, built from the Yelp Open Dataset for VP Analytics. The conclusion is simple to state and harder to earn: the index is defensible as a read of digital-signal health, and it is not, and should not be presented as, a certified measure of operational performance.

**19,154**QSR LOCATIONS  
SCORED, 13,719 OPEN**0.97 →  
0.83**ROC-AUC, NAIVE VS.  
LEAKAGE-AUDITED**4**COMPETING INDEX  
CONSTRUCTIONS  
BUILT AND COMPARED**38.4%**LOCATIONS WHOSE  
RANK SWINGS >15% OF  
THE POPULATION UNDER  
REASONABLE  
ALTERNATIVE WEIGHTS

**What was built.** A thirteen-stage analytical pipeline that scores every location on four pillars (Peer standing, Engagement, Momentum, Customer Experience), tested against three alternative weighting schemes rather than assumed correct in its original form. Customer Experience is measured through a seven-aspect sentiment model covering 1.16 million reviews. Every location also carries a trajectory read, a confidence level, and an intervention priority, alongside a live dashboard that exposes the evidence behind each score rather than just the number.

**What was learned.** A first-pass model predicted open-or-closed status at 0.969 ROC-AUC; that result was leakage, not skill, it was detecting review recency, not health. Removing every recency-dependent feature dropped the honest figure to 0.832, the number this report trusts. A second, independent check found that 38.4% of locations move by more than 15% of the population's rank under reasonable alternative weights, meaning any single location's exact position is far less stable than the aggregate ranking. A third check, run while validating a newly added aspect (Order Accuracy), surfaced and fixed a driver-attribution bug that had been silently mis-assigning the "top negative aspect" for over four thousand locations.

**What can be trusted.** The aggregate ranking: Spearman correlation against LHI-3, the selected construction, stays at 0.95 or higher across five deliberately different weighting scenarios. The leakage-audited predictive signal: 0.832 ROC-AUC from peer standing, engagement, and review sentiment alone, with every circular feature removed. The customer-sentiment evidence: all seven scored aspects move monotonically with star rating despite never seeing the rating during scoring.

**What cannot yet be claimed.** That the index measures operational health rather than digital-signal health, no sales, closure-date, or audit outcome exists in this dataset to test that claim against. That any individual location's rank is stable near a threshold. That most locations have a statistically defensible trend, four in five lack the review history to support one either way. That the framework is ready for automated action, it is not; it is ready for analyst-reviewed decision support.

**What should happen next.** Acquire a real outcome variable (dated closures, sales trend, or pilot audit scores) before the next weighting iteration; govern the priority lookup and every threshold as versioned artefacts, not silent constants; and pilot with an analyst in the loop before any automated workflow depends on this score. Section 12 sets these out in full.



## A repeatable scorecard has to answer four questions an average rating cannot

VP Analytics needs a standardised way to read the health of a large, geographically scattered portfolio without waiting for a site visit or a quarterly review cycle. A single average star rating cannot do this: it collapses location, direction of travel, and evidence strength into one number, and it gives an operator no way to tell a location that is healthy and declining from one that is healthy and stable.

The brief asked for a scorecard, built on the Yelp Open Dataset through a Snowflake-and-Python pipeline, that answers four questions for every outlet:

| Question                          | What a raw average cannot answer                                                          |
|-----------------------------------|-------------------------------------------------------------------------------------------|
| Which locations are at risk?      | A single rating does not separate current state from direction of travel.                 |
| What is driving the risk?         | A rating has no attribution to a pillar, an aspect, or a piece of review evidence.        |
| How do they compare to peers?     | Health only means something against a comparable local peer set, not the whole portfolio. |
| What should be prioritised first? | Health and trajectory combined, not the lowest-scoring location by itself.                |

No sales figure, dated closure, or audit outcome exists anywhere in this dataset. That absence shapes every design decision in this report: every choice below had to earn its place through evidence, because there is no outcome variable to grid-search against and declare a winner. Section 11 returns to this constraint directly; it is the single largest source of uncertainty in the framework, and it is stated here first rather than left for a reviewer to find.



## Thirteen stages, each required to fail loudly before the next could trust it

The analytical layer is thirteen discrete, independently re-runnable stages. Each stage asserts its own correctness before writing output, the same discipline the Snowflake load applies, carried through everything downstream of it. The reason is practical, not procedural: a pipeline that fails silently produces a confident-looking wrong answer, and a scorecard an operator is meant to act on cannot afford that failure mode anywhere in its chain.

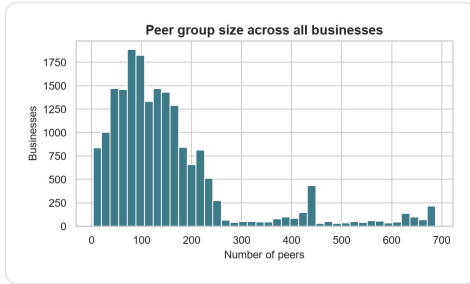
Data QA → Review extraction (1.16M rows) → EDA → Aspect sentiment → Peer / CX scoring → Trajectory → Leakage gate → ML modelling → 4 LHI candidates → Robustness grid → Explainability → Confidence / fallback → Final export

This structure paid for itself twice during the build. The leakage gate (Section 05) caught a model that had learned to detect a closed business rather than an unhealthy one, before that result could shape the index. Independently, validating a newly added aspect (Section 07) surfaced a pre-existing bug in how the "top negative driver" was attributed per location. Neither would have been visible in a single monolithic script; both were visible because each stage was required to assert correctness before the next stage could depend on it.

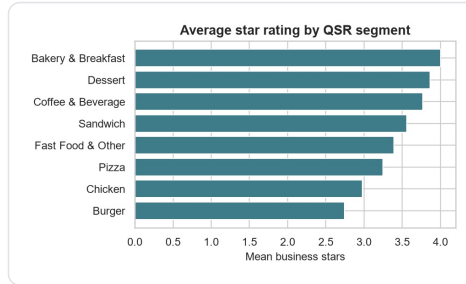


## Most locations have deep peer comparison; a meaningful minority do not

Every location is benchmarked against an adaptive, same-segment peer tier (10 miles, then city, state, global) rather than the whole portfolio, because star ratings vary sharply by segment: a Burger location and a Bakery & Breakfast location are not answering the same question when they are each asked "how healthy is this outlet." The median location has 119 comparable peers. Three hundred and fifteen have fewer than ten, and that thin-peer group is exactly where a peer-relative score deserves the least confidence, which is why peer-tier quality feeds directly into the confidence field in Section 08 rather than being left as an invisible source of noise.



**Peer-group size.** Median 119 comparable peers per location. Low-volume peer groups are where confidence in a peer-relative read is weakest.



**Rating by QSR segment.** Star ratings vary sharply by segment, the direct evidence behind benchmarking a location within its own segment.

Review volume also distorts apparent health, though less cleanly than a textbook story would suggest. Star-rating variance across volume buckets is non-monotonic, not a tidy decline, but average rating does climb steadily with review volume (3.22 to 3.76 stars, smallest to largest bucket), consistent with popularity or survivorship rather than volume mechanically producing more accurate scores. This finding justified shrinking low-volume locations toward a peer prior rather than scoring them on raw averages, without overselling that justification as more than it is: a general statistical safeguard, not a precisely fitted correction.

One further check mattered before any pillar was trusted: whether the four scoring pillars (Peer, Engagement, Momentum, Customer Experience) were genuinely distinct, or secretly the same variable measured four times. Cross-pillar correlation is low once the expected internal overlap inside the Peer pillar is set aside; Engagement and Momentum both sit at a variance-inflation factor of roughly 1.0 to 1.2 against everything else. The four-pillar structure held up under its own scrutiny.



## Leakage inflated apparent model performance; the corrected model retains a credible signal

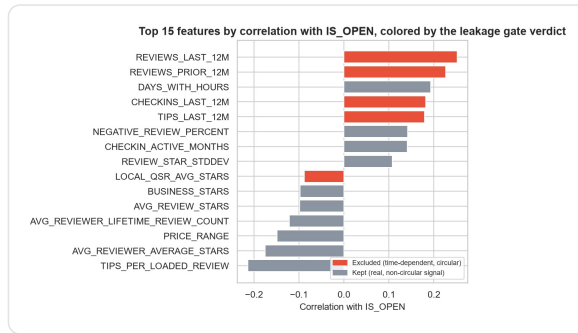
No sales, closure-date, or audit outcome exists in this dataset. In its absence, the nearest available proxy for location health is whether a business is still listed as open, IS\_OPEN. That proxy is dangerous to use naively, and the leakage audit exists to show exactly why, and what survives once the danger is removed.

A closed business does not generate new reviews. Every business in the dataset is measured against one fixed extraction date (19 January 2022), so a location that closed years earlier will trivially show zero recent reviews, zero recent check-ins, zero recent tips, not because those signals reflect declining quality, but because the business stopped existing:

| Metric                                 | Closed Locations | Open Locations |
|----------------------------------------|------------------|----------------|
| Zero reviews in the trailing 12 months | 92.7%            | 12.8%          |
| Median days since last review          | 1,572            | 73             |

Feeding twelve-month review, check-in, and tip activity into a classifier that predicts open-versus-closed status does not validate a health index; it teaches the model to detect its own label.

*The model was not learning what makes a location healthy. It was learning what a closed business looks like in a dataset that stops updating the day it closes.*



**Feature-leakage audit.** The features most correlated with open/closed status, coloured by what the leakage gate decided: high correlation alone was never the reason a feature was excluded.

We ran the same model family twice, once on the full feature set, once after systematically excluding every feature classified as time-dependent or circular, and the gap between the two results is the finding:

**0.969**

ROC-AUC, FULL  
FEATURE SET (NAIVE,  
INCLUDES LEAKY  
RECENCY FEATURES)

**0.832**

ROC-AUC, LEAKAGE-  
AUDITED FEATURE  
SET, THE TRUSTED  
RESULT

**38 / 11**

FEATURES JUDGED  
SAFE / EXCLUDED FOR  
THE TARGET

**0**

MOMENTUM FEATURES  
SAFE FOR THIS TARGET,  
A DISCLOSED GAP

**What 0.832 establishes, and what it does not.** It establishes that peer standing, engagement intensity, and review sentiment carry genuine, moderate predictive value against the proxy, without the recency shortcut that inflated the naive figure. It does not establish that IS\_OPEN is a correct measure of location health, only that it is the least circular target available in this dataset. Every downstream use of this model, including the LHI candidates in Section 06, treats its output as evidence about which features carry signal, never as ground truth the index must match.



## Four constructions were tested against each other; none was assumed correct in advance

Feature importance answers "what predicts this proxy," not "how much should this variable count toward health." Collapsing those two questions would have been the easier way to look data-driven; this report refused to collapse them. Instead, four independently weighted versions of the same four pillars were built and compared, treating the original 30/25/25/20 split as a baseline to test, not an answer to defend:

| Version | Basis                                | Peer  | Engagement | Momentum | Cust. Exp. |
|---------|--------------------------------------|-------|------------|----------|------------|
| LHI-0   | Business-rule baseline               | 30.0% | 25.0%      | 25.0%    | 20.0%      |
| LHI-1   | Statistical, PCA + uniqueness signal | 33.7% | 17.8%      | 14.9%    | 33.5%      |
| LHI-2   | ML-informed, permutation importance  | 30.1% | 43.3%      | 25.0%*   | 1.6%       |
| LHI-3   | Hybrid, synthesis of the three above | 31.8% | 29.2%      | 20.3%    | 18.7%      |

\* Fallback to LHI-0's value; no safe ML feature maps to Momentum (Section 05). Disclosed as a fallback, not fabricated.

LHI-2 is the most striking result in the table: Customer Experience collapses to 1.6% weight, because the safe-feature model finds almost no independent predictive value in the review-percentage proxies available to it. That is disclosed rather than smoothed over. It reflects a real gap between "predicts the proxy" and "matters to an operator," not an error in the construction.

A fifth candidate signal was rejected on evidence, not on instinct

OPERATIONAL\_PROFILE\_DAYS\_WITH\_HOURS, how many days of the week a Yelp listing has posted operating hours, surfaced with a striking permutation importance during model development: rank 6 of 38 safe-set features, ahead of the star rating itself. A weaker analysis would have stopped there and given it a pillar. This report did not, because predictive power answers "does this help predict the proxy," not "should it count toward location health," and those are different questions. The signal was run through an explicit six-step gate instead.

| Check                           | Verdict | Evidence                                                                                                                              |
|---------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------|
| Predicts the proxy outcome      | Yes     | Rank 6/38 by permutation importance (mean = 0.0144)                                                                                   |
| Relationship is stable          | Yes     | Coefficient of variation (std/mean) = 0.068 across repeated permutations                                                              |
| Independent of the four pillars | Yes     | Max  correlation  with an existing pillar = 0.211 (Engagement)                                                                        |
| Adding it preserves the ranking | Yes     | At 10% weight in LHI-3 (rescaling the rest): $\rho = 0.984$ vs. LHI-3, risk-band agreement 86.4%                                      |
| Operationally actionable        | Partly  | Fixable in minutes, but fixes the listing, not the operation                                                                          |
| Actually measures outlet health | Partly  | A listing-completeness signal, not an observation of whether the outlet keeps consistent hours; this dataset cannot see that directly |

Verdict: a contextual diagnostic, not a scored pillar. Five of six checks pass cleanly. The one that fails is the one that matters most by design: a feature can be predictive, stable, independent, and rank-preserving while still not being what "location health" means. It ships as context in the final export, visible to an operator, but it does not move the score. That is a working demonstration that a strong statistical signal can fail a conceptual test even after passing every quantitative one.

**The selection rule was fixed before the results existed**

Not by fiat: the version with the highest mean pairwise rank correlation against the other three, closest to the cross-method consensus, wins.

| Comparison     | Spearman $\rho$ | Risk-Band Agreement |
|----------------|-----------------|---------------------|
| LHI-0 vs LHI-3 | 0.9984          | 95.8%               |
| LHI-1 vs LHI-3 | 0.9778          | 83.8%               |
| LHI-0 vs LHI-1 | 0.9841          | 83.8%               |
| LHI-2 vs LHI-3 | 0.9513          | 81.7%               |
| LHI-0 vs LHI-2 | 0.9378          | 80.1%               |
| LHI-1 vs LHI-2 | 0.8669          | 65.9%               |

LHI-3 wins on its own terms, mean pairwise correlation 0.976 against 0.973 for the baseline it was never guaranteed to beat. Business logic, statistical structure, and predictive importance mostly agree; where they do not (LHI-1 vs LHI-2,  $\rho = 0.87$ ), the hybrid sits closer to the middle than to either extreme. LHI-3 is the version scored and reported throughout the rest of this document.



## Aspect sentiment shows convergent validity, with weaker evidence for rare operational themes

The Customer Experience pillar is built from seven operational aspects scored per review, Food, Service, Staff, Cleanliness, Waiting Time, Order Accuracy, Value, using a keyword taxonomy plus a domain-adjusted VADER lexicon. This is a transparent heuristic, not a contextual language model, and that limitation is disclosed everywhere the aspect scores are used: naive keyword matching can misread contrastive sentences such as "the food wasn't bad but the service was terrible." It was chosen anyway because every match carries an inspectable rule that can be challenged line by line, a property a black-box transformer would not have given this project on a six-week timeline.

Order Accuracy was added after the original six, following a gap review against the brief's named operational themes that found no home for "wrong order" or "gave me the wrong order" in the existing taxonomy. It was run through the same validation every other aspect goes through, rather than assumed useful because it sounded plausible.

| Aspect         | Mentions | Spearman $\rho$ | Monotonic vs. Stars |
|----------------|----------|-----------------|---------------------|
| Service        | 469,539  | 0.623           | Yes                 |
| Cleanliness    | 122,501  | 0.583           | Yes                 |
| Staff          | 305,295  | 0.585           | Yes                 |
| Food           | 976,024  | 0.531           | Yes                 |
| Waiting Time   | 480,368  | 0.421           | Yes                 |
| Value          | 284,549  | 0.408           | Yes                 |
| Order Accuracy | 9,259    | 0.230           | Yes                 |

Convergent validity: aspect sentiment vs. star rating, stars excluded from sentiment construction. Business-clustered bootstrap, all significant at  $q < 0.001$ .

All seven aspects move monotonically with star rating despite never seeing the rating during scoring: convergent validity, not proof, but the right direction of evidence. Coverage reached 94.9% of the 1.16 million reviews, and every location's aspect score is shrunk toward a leave-one-out peer prior, so a business with three reviews cannot look as certain as one with three hundred.

Order Accuracy passes, with a caveat stated plainly rather than smoothed over. Its correlation with star rating is genuine and statistically overwhelming, but noticeably weaker than the other six (0.41 to 0.62). The aggregate signal is real; a business-level trend for it is mostly not. Of the 5,529 locations with any Order Accuracy mentions, 5,526 fall into "Insufficient Evidence" for a recent-versus-prior trend, the same minimum-mentions gate that protects every other aspect's trajectory from noise also, correctly, blocks Order Accuracy's, because the mention volume per location is simply too thin.

### **Validating a new aspect exposed a bug in existing driver-attribution logic**

The bug was found during Order Accuracy's own validation, not after it shipped. The "top negative aspect" shown per location had been selected by peer-percentile alone, with no check for whether the business had any evidence for that aspect. Because Order Accuracy is rare (0.48 mentions per location on average) but almost always negative when mentioned, it won "worst aspect" for 4,686 locations under the old logic, and 74% of those had zero Order Accuracy mentions of their own; the assignment was pure peer-prior shrinkage, not real evidence. This was not a failure mode Order Accuracy introduced, any sufficiently rare, uniformly negative aspect could have triggered it, silently, for as long as the logic existed. The fix now prefers aspects a location has direct evidence for, falling back to the peer-prior-only pick only when a location has no evidence for any aspect at all. After the fix, Order Accuracy legitimately drives 1,212 locations (6.3%), every one backed by a real mention of its own.

Validation itself was a manual spot-check of roughly 180 reviews, not a multi-annotator gold-standard set. That distinction is stated explicitly wherever the aspect scores are used, because it is the honest boundary of what this team could validate against a hard deadline.



## A single score cannot say both how healthy and which direction; we kept them separate

Collapsing health and trajectory into one number produces exactly the failure mode an operator most needs protection from: a good-looking score that is quietly getting worse. Four fields are reported separately instead.

| LHI  | Trajectory    | Meaning                                                                |
|------|---------------|------------------------------------------------------------------------|
| 85 ↑ | Improving     | Healthy and getting better                                             |
| 85 ↓ | Deteriorating | Still above the Healthy threshold (60), but a real decline is underway |
| 50 ↓ | Deteriorating | High priority, below target (Watch band, 40–60) and actively worsening |
| 20 ↓ | Deteriorating | Critical, already in the bottom band and compounding                   |

Trajectory is a recent-versus-prior twelve-month comparison, not a slope drawn through noisy monthly counts, and a location is only labelled Improving or Deteriorating when a bootstrap confidence interval on that shift excludes zero. Everything else is labelled Insufficient Evidence, which is most of the portfolio:

**79.7%**

LOCATIONS WITHOUT  
ENOUGH REVIEW  
HISTORY IN BOTH  
WINDOWS TO  
EVIDENCE A TREND

**357**

STATISTICALLY-  
EVIDENCED  
DETERIORATING  
LOCATIONS

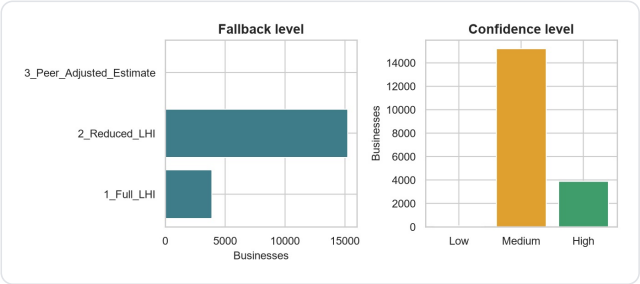
**157**

STATISTICALLY-  
EVIDENCED  
IMPROVING  
LOCATIONS

**3,899**

LOCATIONS WITH REAL  
TRAJECTORY EVIDENCE  
EITHER WAY

Reporting "we don't know" for four in five locations was the deliberate choice over fabricating a trend line to fill a dashboard panel. It is disclosed as a limitation, not hidden inside a smoothed chart.



**Confidence vs. fallback.** Two separately computed measures that agree for all but a handful of locations, nothing about the read is hidden behind one clean-looking score.

**Priority is a lookup, not an arithmetic average**

The first attempt computed priority as risk score plus a trend adjustment, a plausible-looking formula that did not work: on a 0–100 scale, a capped trend nudge of  $\pm 20$  points can never move a genuinely healthy, merely-declining location into the same priority band as a location that is already unhealthy, no matter how the weights are tuned. It was replaced with an explicit health-band  $\times$  trajectory lookup table, verified directly against the live output rather than assumed from the formula, so a Healthy-and-Deteriorating location is always escalated above a Healthy-and-Stable one. Confidence acts as a damper on top of that lookup: a Low-confidence trajectory signal is not allowed to drive urgent action on its own.



## Aggregate rankings are stable; individual locations near a threshold are not

Two different questions get two different answers, and conflating them is the single most common way a scoring model misleads the team that built it.

### Is the ranking stable?

Broadly, yes. Across a five-scenario weight grid (peer-heavy, engagement-heavy, momentum-heavy, CX-heavy, equal-weight) plus outlier winsorisation and an alternate peer-group definition, Spearman correlation against LHI-3, the selected construction, stayed at 0.95–1.00 in every case. Aggregate rank order is not fragile.

### Is any individual location's rank stable?

No, and this is the finding that matters most for how the score gets used. 38.4% of locations move by more than 15% of the entire population's rank across the same six scenarios. A location can sit at the 20th percentile under one reasonable weighting and the 55th under another. The ranking, in aggregate, is trustworthy; any single location's exact position near a threshold is not, which is why a Confidence field ships alongside the score rather than asking anyone to trust one number to two decimal places.

### What does this mean for a threshold-based shortlist?

A list such as "the top 100 at-risk locations" is a policy choice produced by a specific weighting and threshold, not a fixed fact about the portfolio. Given the 38.4% figure above, any such shortlist should always be quoted alongside the settings that produced it, and re-quoted if those settings change. Section 11 records the risk-band cut points themselves (60/40/25) as a further, separate source of this same sensitivity: they are round, business-interpretable choices, not statistically fitted ones.

### Does the leakage-audited model still say anything real?

Yes. 0.832 ROC-AUC against a 0.5 random baseline, using only features that cannot trivially encode the target. This report did not require that number to be low just because the leaky version was inflated; a model that genuinely predicts open-or-closed status from peer standing and review sentiment, without cheating, is a legitimate and separately useful finding, distinct from the claim that the score measures operational health.



## The dashboard is built to disclose evidence, not just display a score

The analytical layer feeds a live executive dashboard covering the 13,719 open locations: portfolio KPIs, a Deteriorating count with an early-warning callout for currently-healthy locations, a geographic risk view, an aspect-sentiment breakdown, a sortable location table, and a per-location detail panel that walks the full diagnostic chain, LHI → weakest pillar → weakest aspect → representative review evidence → recommended operator focus. Every number in that panel, including the pillar-to-LHI point contributions, reconciles exactly to the displayed score, so an operator questioning the number can trace it rather than take it on faith.

### 36.9%

OPEN LOCATIONS  
RATED HIGH OR  
CRITICAL RISK

### 347 / 268

DETERIORATING  
LOCATIONS FLAGGED,  
268 CURRENTLY  
HEALTHY OR WATCH

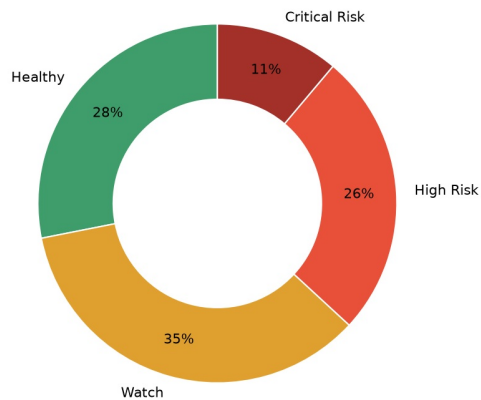
### 48.4

PORTFOLIO-AVERAGE  
LHI (0–100 SCALE, ALL  
19,154 LOCATIONS)

### 1

SELF-CONTAINED HTML  
FILE, NO SERVER, NO  
EXTERNAL MAP TILES

Portfolio risk distribution, 13,719 open locations



**Portfolio risk mix.** 36.9% of the 13,719 open locations are currently rated High or Critical Risk.



## What would not survive a sceptical review, stated first

A report earns trust by surviving the questions it did not have to raise itself. The assumptions below are choices this project made deliberately; the limitations are gaps it did not close. Both are stated once, here, rather than scattered as footnotes.

### Assumptions

| Assumption                   | Detail                                                                                                                                                                                                                                                                              |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Target proxy</b>          | No sales, dated closure, or audit outcome exists in this dataset. IS_OPEN, with leaky features removed, is used as evidence about signal quality, never as ground truth the index must match.                                                                                       |
| <b>Fixed extraction date</b> | Every "recent" or "prior" window is anchored to 19 January 2022, not to each business's own history. Correct for a historical snapshot; would need re-anchoring for a live feed.                                                                                                    |
| <b>Peer comparability</b>    | Peers are same-segment, geographically adaptive (10mi → city → state → global). 99.4% of locations resolve to the tightest, most local tier.                                                                                                                                        |
| <b>Aspect taxonomy</b>       | Seven aspects, extended once, on evidence (Order Accuracy). The new aspect went through the same convergent-validity and monotonicity checks as the original six, and its weaker, trend-unreliable result is disclosed rather than smoothed into "seven aspects validated equally." |
| <b>Operational profile</b>   | DAYS_WITH_HOURS passed five of six evidence-gate checks (Section 06) but was kept out of the LHI because it measures listing completeness, not observed operating-hours consistency. A disclosed judgment call, not a statistical result.                                           |
| <b>Hybrid weights</b>        | LHI-3 averages the other three candidates by a documented, simple rule. It was not grid-searched against an outcome, because no defensible outcome exists to search against.                                                                                                        |
| <b>Risk-band thresholds</b>  | Healthy/Watch/High Risk/Critical Risk are fixed LHI values (60/40/25), not fixed population shares, so the portfolio's aggregate health can be seen moving over time. The specific numbers are round, business-interpretable choices, not statistically derived.                    |

## Limitations

- **QSR classification is a rule-based approximation**, carried forward rather than re-audited. The 19,154-business population was defined upstream by category-token matching plus a table-service and geography filter. By that filter's own accounting, 77% of retained locations carry no parseable table-service flag; a follow-up layer classified 9,238 as "Supported" QSR, 6,592 as "Uncertain," and 3,324 flagged for manual review. A small number of independent full-service restaurants may therefore be misclassified as QSR, and every score in this report carries that risk forward rather than correcting for it.
- **The data is historical.** Latest review: 19 January 2022. Nothing here describes 2026 operating conditions.
- **Yelp's user base is not a representative customer sample** for any given operator's actual footfall, and the dataset's 13-state, 8-segment coverage is a Yelp artefact, not a sampling design.
- **The sentiment layer is a lexicon, not a language model:** inspectable and reproducible, weaker on sarcasm, negation, and contrastive clauses than a modern transformer would be.
- **Order Accuracy is validated at the aggregate level, not the per-location trend level.** 5,526 of 5,529 locations with any Order Accuracy mentions have too few to support a recent-versus-prior trend; the static score is real evidence, a per-location trajectory for it mostly does not exist yet in this dataset.
- **Governance is not yet in place.** Every chart and number here is code-level reproducible, but the lexicon, taxonomy, thresholds, and weighting scenarios are not yet a single versioned artefact that a reviewer could audit independently of this report.
- **No franchise or multi-site ownership resolution.** Chain-level roll-ups were explicitly out of scope; each row is one physical location.
- The 38.4% rank-reversal figure (Section 09) is a property of any score built this way, not a defect specific to this construction, but it means a shortlist generated at one threshold setting should never be presented as the definitive list without restating the setting that produced it.



## Five evidence-led steps before this framework earns operational trust

None of the recommendations below assume the framework is ready for automated action. It is ready for analyst-reviewed decision support; the steps here are what would close the gap toward more autonomous use, and they are ordered by how directly each one addresses the largest source of uncertainty already named in this report, the absence of a real outcome variable.

### Strengthen the Evidence

- 1 Acquire a real outcome variable before the next weighting iteration. Dated closures, sales trend, or audit scores from even a handful of pilot franchise groups would let LHI-2's weighting, and every threshold in Section 11, mean something beyond "predicts our own proxy."
- 2 Re-run the sentiment validation with a multi-annotator gold set. The lexicon approach is a sound first version; a small labelled set (500 reviews, three annotators) would establish whether moving to a fine-tuned transformer is justified by the accuracy gain.

### Govern the Framework

- 3 Ship the priority lookup table as a governance artefact, not just code. Operators trust a rule they can read over a black-box score; the health × trajectory table in Section 08 is designed to be shown, not hidden.
- 4 Version every fixed-threshold shortlist as a policy choice. Given the 38.4% rank-reversal finding, "the top 100 at-risk locations" should always be quoted with the weight and threshold settings that produced it, and re-quoted when those settings change.

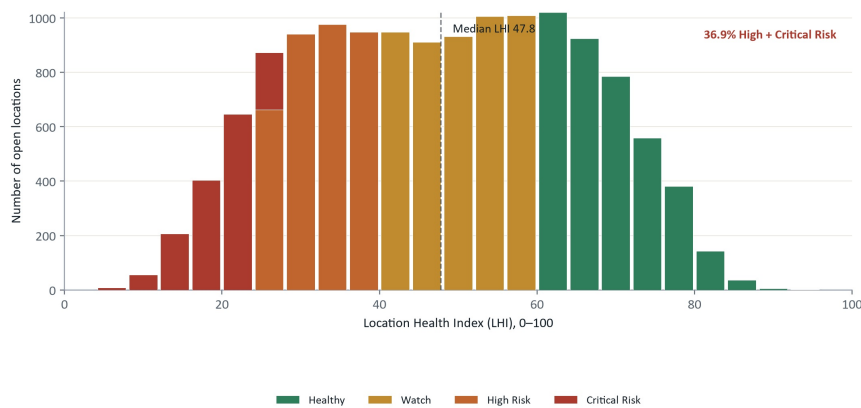
### Deploy Responsibly

- 5 Pilot with an analyst in the loop before automating. The right next step is review of the flagged locations against real operational knowledge, not wiring the priority queue directly into an intervention workflow.
- 6 Audit every low-volume evidence-selection rule for the same shrinkage bias Order Accuracy exposed. The driver-attribution fix in Section 07 was found by validating a new aspect, not by auditing the six already in production, meaning it went undetected there for as long as the logic existed. Any other selection rule that picks a "worst" or "best" category from unevenly-sized evidence pools should be checked for the same failure mode before it is trusted.



FIGURE 07

## Risk is concentrated in a sizeable minority of QSR locations

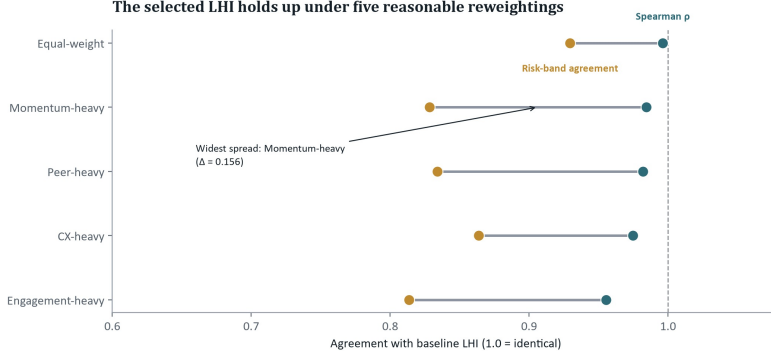


Source: QSR\_FINAL\_LHI\_DATA.csv × QSR\_DRIVERS.csv (IS\_OPEN), analysis/output/, n = 13,719 open locations, 4-point bins.  
Colour composition per bin reflects each location's true RISK\_CATEGORY, not the bin midpoint — 1 of 25 bins straddle the 25/40/60 threshold.  
Risk-band-composition histogram, not a generic "stacked histogram" — no such R Graph Gallery pattern exists (see design spec).

**Score distribution.** Risk is concentrated in a sizeable minority of locations, not spread evenly across the portfolio; the median open location scores 47.8.

FIGURE 09

## The selected LHI holds up under five reasonable reweightings

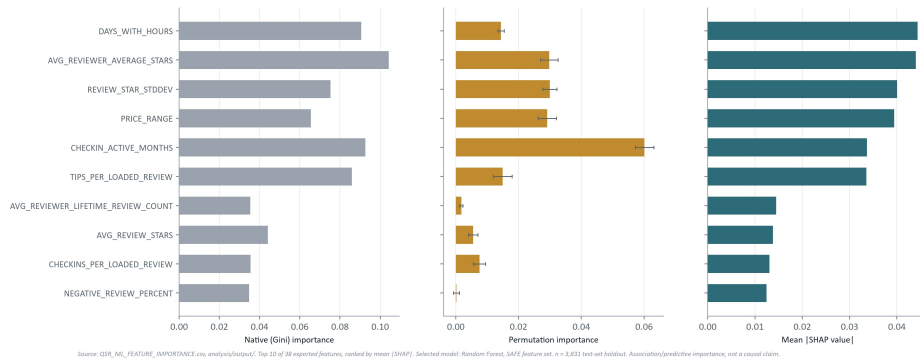


Source: QSR\_LHI\_SENSITIVITY.csv, analysis/output/, n = 19,154 classified businesses, scored under each weighting scenario.  
Dumbbell connects each scenario's two agreement metrics vs. the selected construction (LHI-3, the hybrid scored throughout this report), baseline\_LHI3 (=1.0 by definition) omitted as a row since it carries no information. Sorted ascending by Spearman  $\rho$ .

**Robustness under alternative weightings.** The selected construction (LHI-3) agrees with five deliberately different reweightings at 0.95 Spearman correlation or higher in every case.

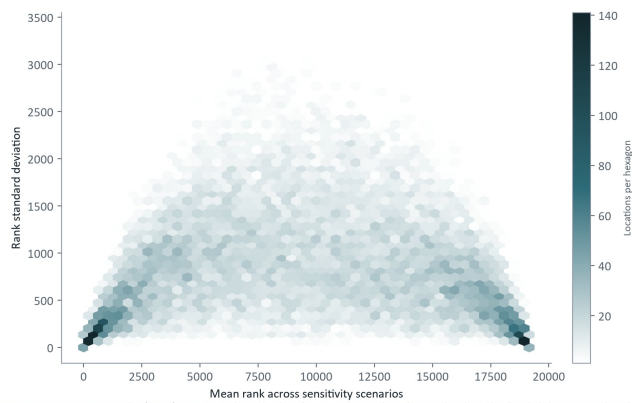
**Reviewer credibility and rating consistency drive the closure-risk signal**

FIGURE 48



**What drives the leakage-audited model.** Three independent importance measures (native, permutation, SHAP) agree on the same top features, none of which trivially encode the target.

**Mid-pack locations swing more than the extremes**



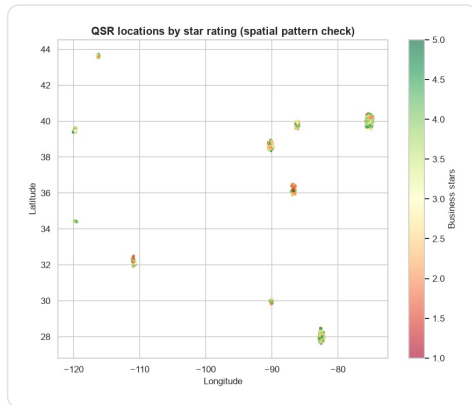
**Rank stability by position.** Locations near the top or bottom of the portfolio barely move across weighting scenarios; it is the mid-pack locations whose rank swings most, hexagon-binned to avoid overplotting at  $n = 19,154$ .

Early exploratory analysis

The four charts below are working diagnostics from the earliest stage of the pipeline, kept here in their original form rather than restyled, to show the process that shaped every later decision.



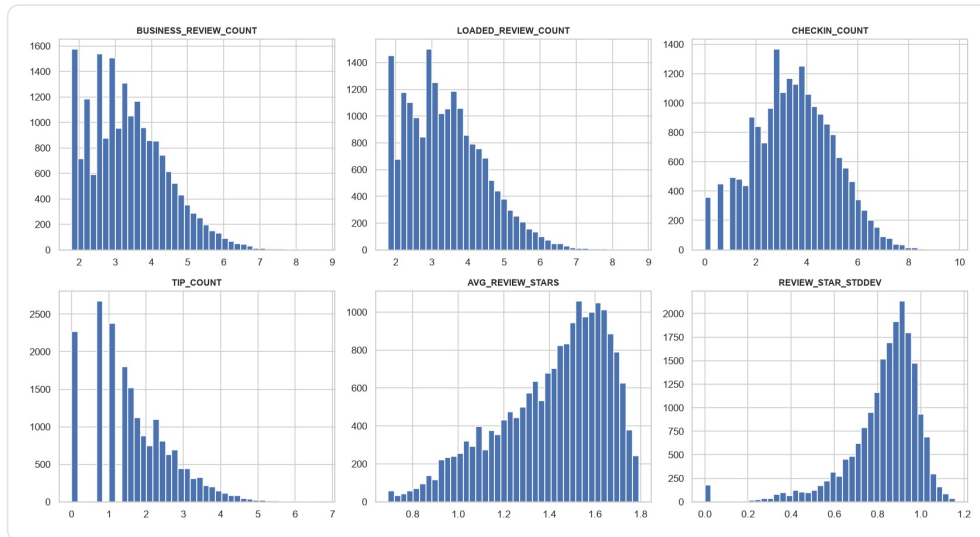
**Aspect sentiment vs. star rating, first pass.** An early convergent-validity check, before the final aspect-sentiment model was built: all seven aspects trend upward with star rating, and Order Accuracy is already visibly the weakest, flattest line, a signal that held up all the way through to the final model.



**Geographic coverage check.** QSR locations cluster into a handful of metro areas rather than spreading evenly across the country, the direct evidence behind treating the dataset's geographic coverage as a Yelp artefact, not a designed sample.



**Review volume over time.** Volume grows steadily for over a decade, then drops sharply from 2020, the visual evidence behind treating this dataset as a historical snapshot rather than a live feed.



**Core field distributions, first pass.** An early shape-and-outlier check on review counts, check-ins, tips, and star-rating statistics, run before any feature engineering, to catch data-quality problems before they could propagate downstream.

Prepared by Sharath Sista for the VP Analytics QSR Location Health Index project, LSE Data Analytics Career Accelerator. Source data: Yelp Open Dataset, historical through 19 January 2022. No claim in this report describes current operating conditions. All figures are reproducible from the accompanying notebook and pipeline source.